

ArTE-VLA: Autoregressive Target Effects for Vision-Language-Action Models

Anonymous Authors

Abstract—Vision-language-action (VLA) models typically generate robot actions directly from visual observations and language instructions, leaving the intended physical evolution of the scene implicit. We introduce ArTE-VLA, a VLA that predicts how the language-referred target should change before generating an action. Given dual-view RGB observations and an instruction, ArTE-VLA infers an implicit target representation and autoregressively predicts target effects at four future horizons. Each effect describes translation, rotation, linear and angular velocity, and activity. The predicted effects are embedded as tokens and appended to the visual-language prefix, allowing the action expert to condition on a compact physical future. Simulator states provide effect supervision during training but are not required at inference. Across three evaluation seeds on all ten LIBERO-Object tasks, ArTE-VLA improves the success rate of a recipe-matched P10.5 baseline from 70.9% to 84.3%, a 13.4 percentage-point gain over 1,500 paired episodes per method. Fixed-noise interventions further show that changing the predicted effect values produces consistent changes in generated actions. These results suggest that a compact, target-centric future representation can provide an effective and inspectable interface between VLA perception and control.

I. INTRODUCTION

Vision-language-action models have emerged as a promising route toward general robot policies [1], [2], [3], [4]. By combining pretrained visual-language representations with large collections of robot demonstrations, recent VLAs can follow open-vocabulary instructions and transfer across objects, tasks, and environments [5], [6], [1]. Most existing architectures, however, are optimized to map the current observation directly to an action sequence [7], [8]. The physical outcome that connects an instruction to an action—which object should move, in what way, and over what time scale—remains latent inside the policy.

This direct formulation creates two related difficulties. First, the action expert must simultaneously resolve language grounding, infer task-relevant dynamics, and generate precise controls [9], [10]. Second, the resulting computation is difficult to inspect: a successful rollout does not reveal whether the policy anticipated the intended object motion or relied on a less robust correlation [9], [11]. Predictive world models and visual chain-of-thought methods address this problem by forecasting future images or point trajectories before acting [12], [13], [14], [15]. Although expressive, these intermediate representations can be high-dimensional and need not isolate the physical state of the object that is central to the instruction [16].

We propose ArTE-VLA, which inserts a compact target-effect bottleneck into a pretrained VLA. Rather than predicting a complete future image, the model compresses its

visual-language prefix into an implicit representation of the primary target and predicts the target’s physical effects at four horizons. Each prediction contains a 3-D translation, a 3-D rotation, linear and angular velocity, and an activity logit. The predictions form an autoregressive chain: the effect at one horizon is used to predict the next. Horizon-specific hidden states and effect values are then projected into tokens that condition the flow-matching action expert.

The target-effect representation is learned with privileged supervision but used without privileged inputs. We recover object states from the simulator states stored in LIBERO demonstrations and construct aligned multi-horizon effect labels. These labels appear only in the auxiliary prediction loss; the action expert always receives model-predicted effects. At test time, ArTE-VLA uses the same RGB observations, instruction, and robot state as vanilla P10.5. This separation allows structured physical supervision to shape the policy without introducing an object detector, simulator state, task identifier, or ground-truth target state into the deployed model.

Our main contributions are:

- We introduce a compact, target-centric physical-effect bottleneck for VLA policies. The bottleneck predicts a four-horizon autoregressive future and directly conditions the action expert.
- We develop a replay-based labeling pipeline that adds aligned target effects to robot demonstrations while preserving their original images, actions, states, and language instructions.
- We formulate value-preserving interventions that separate reliance on the complete effect token from reliance on its explicit physical values and temporal order.
- On LIBERO-Object, ArTE-VLA improves a matched P10.5 baseline by 13.4 percentage points across three evaluation seeds and 1,500 paired episodes per method, while revealing substantial variation across target objects.

II. RELATED WORK

A. Vision-Language-Action Models

VLA models transfer pretrained vision-language priors to robot control through diverse manipulation trajectories. Octo supports language or goal-image conditioning and adaptation to new observation and action spaces [17], while OpenVLA emphasizes an open, scalable VLA that can be fine-tuned for downstream manipulation [18]. Flow-matching policies avoid discretizing continuous controls: π_0 couples a vision-language backbone to a continuous action expert [19], and

PI0.5 extends this family with heterogeneous co-training for open-world manipulation [5]. These systems primarily map the current multimodal context directly to actions. ArTE-VLA uses PI0.5 as its backbone but adds a jointly supervised physical prediction interface that is consumed by the otherwise unchanged action expert.

B. Predictive Representations for Control

Predictive representations make intended scene evolution explicit before or alongside action generation. GR-2 jointly learns video and action prediction from web videos and robot trajectories [20]; CoT-VLA generates future visual goals before actions [12]; and WorldVLA places image and action generation in a shared autoregressive formulation [13]. These visual futures retain appearance and multi-object context, but also encode scene content irrelevant to the commanded manipulation. ArTE-VLA neither synthesizes future observations nor performs explicit planning with learned dynamics. It predicts a low-dimensional physical change for one language-conditioned target and inserts it into the action context. Thus, visual prediction preserves rich scene evolution, whereas target effects expose compact values whose temporal order and influence on actions can be directly tested.

C. Object and Point Motion as Policy Guidance

Object-centered conditioning separates task-relevant entities from the scene. MOO extracts object cues from the image and command to support manipulation of unseen categories [21], but describes the current target rather than its desired evolution. Motion representations complement this grounding: Any-point Trajectory Modeling forecasts arbitrary image-point tracks for policy learning [14], and Track2Act converts predicted goal-conditioned tracks into object and end-effector motion plans [15]. ArTE-VLA instead predicts one instruction-grounded target’s multi-horizon 3-D translation, rotation, velocities, and activity, then injects these effects as tokens into an end-to-end VLA action expert. Because its target representation is implicit, it is not a standalone detector or calibrated tracker; the structured effects serve as a compact supervision and action-conditioning bottleneck.

III. METHOD

A. Problem Formulation

At time t , the policy receives workspace and wrist RGB observations $I_t = (I_t^{\text{ws}}, I_t^{\text{wr}})$, proprioceptive state s_t , and a language instruction ℓ . As in PI0.5, the normalized state is discretized and included in the language prompt, while both images are encoded into visual tokens. Let $X_t = \Phi(I_t, s_t, \ell)$ denote the resulting visual-language prefix. A conventional VLA directly models an action chunk $A_t = (a_t, \dots, a_{t+C-1})$ from X_t . We instead introduce an intermediate target-effect chain $E_t = (e_t^1, \dots, e_t^H)$ and use the

factorization

$$p(A_t, E_t \mid I_t, s_t, \ell) = \left[\prod_{h=1}^H p_\phi(e_t^h \mid e_t^{<h}, X_t) \right] \times p_\theta(A_t \mid X_t, T(E_t)), \quad (1)$$

where $T(E_t)$ embeds the predicted effects into the PI0.5 prefix space. We use $H = 4$ offsets $k_h \in (1, 4, 8, 16)$, corresponding to $(0.1, 0.4, 0.8, 1.6)$ s at the 10-Hz control rate. The intermediate chain asks the model to first represent how the instructed target should evolve and then generate the complete action chunk conditioned on that evolution. The autoregression in Eq. (1) is a temporal factorization; it does not identify a structural causal model.

B. Implicit Target Representation

The backbone produces $X_t \in \mathbb{R}^{N \times d}$, including visual patches from both cameras and tokens from the instruction-state prompt. With prefix validity mask m_i , we form a global context $\bar{x}_t$ and predict one compact target latent:

$$\bar{x}_t = \frac{\sum_{i=1}^N m_i X_{t,i}}{\sum_{i=1}^N m_i}, \quad z_t = f_{\text{target}}(\bar{x}_t) \in \mathbb{R}^{20}. \quad (2)$$

Here f_{target} is a layer-normalized two-layer MLP with a 256-D hidden layer. The vector z_t is inserted as the sole valid slot of the effect predictor; the remaining slots in the shared multi-entity implementation are masked. Because z_t is inferred jointly from vision, language, and state, it can represent the primary object or articulated part referred to by the command without an explicit detector or object identifier. Importantly, z_t is not regressed to the current simulator state. Its semantics arise only through the downstream future-effect and action objectives, so we call it an *implicit target representation*, not a calibrated 20-D physical state.

C. Multi-Horizon Effect Chain

The effect chain first combines the target latent with the pooled backbone context,

$$g_t = \text{Enc}_E(f_z(z_t) + f_x(\bar{x}_t)). \quad (3)$$

All effect hidden states are 256-D; Enc_E and Refine_E are two-layer, four-head Transformer encoders. The latter is shared across horizons and applied recursively. With $\hat{e}_t^0 = \mathbf{0}$ and a learned horizon embedding u_h ,

$$c_t^h = \text{Refine}_E(g_t + W_e \hat{e}_t^{h-1} + u_h), \quad (4)$$

$$\hat{e}_t^h = f_E(c_t^h). \quad (5)$$

Thus, a later prediction receives the model’s earlier prediction rather than a ground-truth previous effect. Each output has the fixed schema

$$\hat{e}_t^h = [\Delta p_t^h, \Delta r_t^h, v_t^h, \omega_t^h, q_t^h] \in \mathbb{R}^{13}, \quad (6)$$

where $\Delta p_t^h \in \mathbb{R}^3$ and $\Delta r_t^h \in \mathbb{R}^3$ describe translation and axis-angle rotation from t to $t + k_h$, $v_t^h, \omega_t^h \in \mathbb{R}^3$ are linear and angular velocities at $t + k_h$, and q_t^h is an activity logit. Translation and velocity are expressed in the end-effector frame at t ; the rotation increment is the logarithmic map

of $R_t^\top R_{t+k_h}$. This target-relative temporal schema separates intended physical change from the robot controls used to produce it.

D. Conditioning the Action Expert

The 13-D values alone do not encode all visual and task context needed by the action expert. We therefore concatenate each prediction with its horizon-specific hidden state and project it to the backbone width:

$$\tau_t^h = f_{\text{token}}([c_t^h; \hat{e}_t^h]) \in \mathbb{R}^d. \quad (7)$$

In our PI0.5 backbone, $d = 2048$. The four effect tokens are appended to the original prefix, $\tilde{X}_t = [X_t; \tau_t^1; \dots; \tau_t^4]$, and are visible to the flow-matching action expert. The original prefix is never replaced, so the expert retains direct access to the images, instruction, and state in addition to the predicted future. Moreover, supervision-validity masks are used only in the effect loss: all four predicted tokens remain present during action training and inference. This prevents episode-boundary information from leaking into the policy through a label-derived token mask.

For completeness, let $\epsilon \sim \mathcal{N}(0, I)$ and $\gamma \in (0, 1)$ be the action-flow noise and sampled flow time. Training constructs $A_t^\gamma = (1 - \gamma)A_t + \gamma\epsilon$ with target velocity $u_t = \epsilon - A_t$. The action expert predicts $v_\theta(A_t^\gamma, \gamma | \tilde{X}_t)$ and minimizes

$$\mathcal{L}_{\text{action}} = \mathbb{E}_{A_t, \epsilon, \gamma} \left[\left\| v_\theta(A_t^\gamma, \gamma | \tilde{X}_t) - u_t \right\|_2^2 \right]. \quad (8)$$

At inference, numerical integration starts from Gaussian noise and follows the predicted vector field from $\gamma = 1$ to 0, yielding the complete action chunk rather than a residual correction to a separately generated action.

E. Training Objective

For each valid horizon, simulator replay supplies a continuous target $y_{t,h}^{1:12}$, activity label $y_{t,h}^{\text{act}}$, and validity mask $M_{t,h}$. Let $K_t = \max(1, \sum_h M_{t,h})$. The two effect losses are

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_t \left[\frac{1}{K_t} \sum_h M_{t,h} \frac{1}{12} \sum_{j=1}^{12} \rho(\hat{e}_{t,h}^j - y_{t,h}^j) \right], \quad (9)$$

$$\mathcal{L}_{\text{act}} = \mathbb{E}_t \left[\frac{1}{K_t} \sum_h M_{t,h} \text{BCEWithLogits}(q_t^h, y_{t,h}^{\text{act}}) \right], \quad (10)$$

where ρ is Smooth- L_1 . The full end-to-end objective is

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_E (\mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{act}}), \quad (11)$$

with $\lambda_E = 1$. The reported model uses the raw effect objective: continuous components, active and inactive targets, and positive and negative activity labels receive unit weight before the valid-horizon normalization above. Both losses update the model jointly. Crucially, there is no teacher forcing in the effect recursion, and the action expert is always conditioned on predicted effects rather than replay labels. Training and deployment therefore share the same target-to-effect-to-action path.

F. Replay-Based Effect Supervision

We obtain structured supervision without changing the policy’s deployment inputs. For every demonstration frame, the enrichment procedure restores the stored flattened MuJoCo state and calls the simulator’s kinematic forward pass; it never steps the recorded action. Matching stored states at t and $t+k_h$ gives, with $R_{\text{ee},t}$ denoting the current end-effector orientation,

$$\Delta p_t^h = R_{\text{ee},t}^\top (p_{t+k_h} - p_t), \quad (12)$$

$$\Delta r_t^h = \text{Log}(R_t^\top R_{t+k_h}), \quad (13)$$

$$v_t^h = R_{\text{ee},t}^\top v_{t+k_h}, \quad (14)$$

$$\omega_t^h = R_{\text{ee},t}^\top \omega_{t+k_h}. \quad (15)$$

The target is marked active if translation exceeds 1 mm, rotation exceeds 1° , linear velocity exceeds 0.01 m/s, or angular velocity exceeds 0.05 rad/s. A horizon is valid only when the same target exists at both frames and $t+k_h$ remains inside the episode.

Object names and simulator body identifiers are retained only as enrichment provenance. The enriched data preserve the original RGB observations, proprioception, actions, instructions, and frame indices. During training, the task index selects the teacher target from the multi-entity replay tensor and packs its labels into the single active target slot. In LIBERO-Object the target is slot 0 for all ten tasks. Neither the task index nor the teacher slot, object identity, effect label, or simulator state is embedded by the policy. At deployment, the model uses only the same images, instruction, and robot state as the PI0.5 baseline.

G. Effect Interventions

To test what information reaches the action expert, we intervene after effect prediction while fixing the observation, language, checkpoint, initial action noise, and flow-integration schedule. *Mask* invalidates all effect tokens, measuring dependence on the added pathway as a whole. *Zero-12D* sets only the 12 continuous values in Eq. (6) to zero while preserving each horizon context c_t^h and activity logit. *Reverse* assigns the four 13-D predictions in reverse temporal order, and *shuffle* substitutes predictions from another sample. For value interventions, the altered effects are re-encoded with the receiving sample’s original hidden contexts; its visual, language, and current-state information is never swapped. The resulting tests separate dependence on the token pathway from sensitivity to explicit physical values and their temporal organization.

TODO: Add the main architecture figure showing the PI0.5 backbone, implicit target representation, four autoregressive effect steps, action conditioning, and the training-only simulator-supervision pathway.

IV. EXPERIMENTS

A. Experimental Setup

1) *Benchmark and Data:* We evaluate on LIBERO-Object, which contains ten language-conditioned tabletop manipulation tasks [22]. The training set contains 454 available demonstration episodes and 66,984 frames. Replay

enrichment yields 253,423 valid target-horizon labels, of which 170,107 are active. We verify frame-level alignment between the original and enriched datasets and observe a maximum action-alignment error of 2.90×10^{-8} . Terminal horizons without a valid successor state are masked.

2) *Training Setup*: Both vanilla PI0.5 and ArTE-VLA are initialized from `lerobot/pi05.base` and fine-tuned for 30,000 updates using eight NVIDIA H20 GPUs, global batch size 64, bfloat16 precision, and seed 1000. We use the same optimizer, learning-rate schedule, image resolution, action chunk length of 16, and five executed actions per inference call. ArTE-VLA adds the target head, effect chain, effect-token projection, and the auxiliary loss in Eq. (11); the remaining training recipe is unchanged.

TODO: Add the exact optimizer, learning rate and schedule, weight decay, image resolution, augmentation, checkpoint-selection rule, wall-clock training cost, and trainable-parameter overhead for reproducibility.

3) *Evaluation Protocol*: For each task, we evaluate both policies from the same 50 demonstration-derived initial simulator states under evaluation seeds 1000, 2000, and 3000. This produces 500 paired episodes per method and seed, or 1,500 episodes per method overall. All three evaluations reuse the same seed-1000 training checkpoints; they measure stochastic rollout variability rather than variability across independent training runs. Rollouts use synchronous batches of ten environments, ten flow-matching inference steps, and a fixed time horizon. Success is recorded without terminating the episode early. We report per-task and aggregate success, paired rescue and regression counts, and the exact two-sided McNemar test. Because these resets reconstruct expert demonstration states rather than the benchmark’s standard initial-state set, we keep this protocol separate from comparisons to previously published LIBERO results.

4) *Baselines and Ablations*: The primary baseline is recipe-matched vanilla PI0.5. To distinguish future prediction from auxiliary regularization and added capacity, the complete study also considers an effect predictor that does not condition actions, learned context tokens without structured values, current-target-state conditioning, non-autoregressive horizon prediction, and a privileged ground-truth-effect oracle. We evaluate removal, zero-value, reverse-order, and cross-sample shuffle interventions on the same ArTE-VLA checkpoint. These comparisons isolate three questions: whether the additional pathway improves control, whether future structure matters beyond additional tokens, and whether the trained action expert uses the predicted values.

TODO: Run and report every baseline and ablation named above, including capacity-matched and value-free controls. Remove any comparison from the prose if it is not completed.

5) *Metrics*: Closed-loop success is the primary metric. For effect prediction, we measure translation average and final displacement error, rotation error, linear and angular velocity error, and activity precision and recall. We also report prediction improvement over a zero-motion baseline. Intervention effects are measured by the fraction of changed action chunks, maximum absolute action difference, and relative root-mean-square (RMS) distance. Efficiency is mea-

sured with trainable parameter count, inference latency, and peak accelerator memory.

TODO: Add the effect-prediction and efficiency results promised by this section, with per-horizon errors, a zero-motion comparison, parameter overhead, latency, and peak memory; otherwise remove unreported metrics.

B. Closed-Loop Performance

Table I compares ArTE-VLA with vanilla PI0.5 and reserves matched columns for the auxiliary-only and value-free-token baselines. Across the three evaluation seeds, target-effect conditioning improves mean success from 70.9% to 84.3%, an absolute gain of 13.4 percentage points. The gains for seeds 1000, 2000, and 3000 are 12.8, 14.8, and 12.6 points, respectively. Pooling the 1,500 paired rollouts, ArTE-VLA produces 380 rescues and 179 regressions. The corresponding exact two-sided McNemar test yields $p = 1.20 \times 10^{-17}$.

TODO: Repeat the full training with independent initialization seeds. Add a separate standard-LIBERO-initialization evaluation; do not merge it with the expert-demonstration-reset protocol used in Table I.

The aggregate improvement is dominated by two large rescues. On Butter and Chocolate Pudding, vanilla PI0.5 achieves only 4.7% and 14.7%, whereas ArTE-VLA reaches 90.0% and 86.7%, respectively. The gain is therefore not a uniform shift across tasks. Ketchup and Milk decrease by 33.3 and 27.3 points, respectively, and regress under every evaluation seed. BBQ Sauce decreases by 5.3 points but includes one tied seed. Salad Dressing also changes sign across seeds. This heterogeneity indicates that target effects can resolve severe failures of a direct policy, but can also introduce task-dependent interference.

C. Component Ablations

Table II isolates the contribution of action conditioning, structured effect values, present-versus-future target information, and autoregressive horizon modeling. Each learned variant will use the same initialization, training budget, and evaluation protocol as the full model.

TODO: Populate Table II with separately trained, recipe-matched checkpoints and independent training-seed replications.

D. Action Sensitivity Analysis

We perform a fixed-noise action audit on 32 observations drawn equally from Butter and Chocolate Pudding. Repeating the unmodified forward computation gives zero numerical action difference. Each value intervention changes the action chunk for all 32 samples. Zeroing the 12 continuous effect values yields a median maximum absolute action difference of 0.00956 and a median relative RMS distance of 0.943%. Reversing the horizon order increases these medians to 0.01198 and 1.109%, while cross-sample shuffling yields 0.01277 and 1.306%.

These results establish computational dependence of the action expert on the predicted physical values. They do not establish that those values alone account for the closed-loop improvement: each effect token also contains a learned horizon context, and the zero-value median lies slightly below the prespecified 1% relative-RMS decision threshold. Closed-loop interventions over the full task distribution are

TABLE I

LIBERO-OBJECT SUCCESS (%) ACROSS THREE EVALUATION SEEDS USING FIXED SEED-1000 TRAINING CHECKPOINTS. ENTRIES ARE MEANS ACROSS SEEDS; EACH SEED CONTAINS 50 PAIRED EPISODES PER TASK. BLANK CELLS INDICATE MATCHED BASELINES AWAITING TRAINING AND EVALUATION.

Target object	PI0.5	Auxiliary-only	Value-Free Tokens	ArTE-VLA	Δ vs. PI0.5
Alphabet Soup	77.3			74.7	-2.7
Cream Cheese	74.0			96.0	+22.0
Salad Dressing	86.7			87.3	+0.7
BBQ Sauce	93.3			88.0	-5.3
Ketchup	97.3			64.0	-33.3
Tomato Sauce	70.7			90.0	+19.3
Butter	4.7			90.0	+85.3
Milk	95.3			68.0	-27.3
Chocolate Pudding	14.7			86.7	+72.0
Orange Juice	95.3			98.7	+3.3
Overall	70.9			84.3	+13.4

TABLE II

MATCHED COMPONENT ABLATIONS ON LIBERO-OBJECT. BLANK CELLS DENOTE EXPERIMENTS THAT HAVE NOT YET BEEN COMPLETED.

Variant	Success	Δ vs. PI0.5
Auxiliary-only		
Value-Free Tokens		
Current-target conditioning		
Non-autoregressive effects		
Full ArTE-VLA	84.3%	+13.4

TABLE III

FIXED-NOISE EFFECT INTERVENTIONS ON 32 OBSERVATIONS. ACTION METRICS ARE MEASURED RELATIVE TO THE UNMODIFIED COMPUTATION.

Intervention	Intervention content	Max abs. median	Rel. RMS median (%)	Changed (%)
Repeat	Identical-condition repeat	0	0	0
Zero-12D	Zero 12-D future-state values	0.00956	0.943	100
Reverse	Reverse values across horizons	0.01198	1.109	100
Shuffle	Cross-sample substitution	0.01277	1.306	100

therefore necessary to quantify the independent control value of the structured prediction.

TODO: Extend the value intervention beyond the current 32 observations and two tasks. Run all-task action audits and closed-loop mask, zero-value, reverse-order, and cross-sample-shuffle interventions before making an independent value-use claim.

E. Task-Level Analysis

The paired outcomes suggest two complementary failure modes. For tasks on which vanilla PI0.5 rarely succeeds, target-effect supervision may provide an additional object-motion signal that disambiguates action generation. Conversely, the regressions on Ketchup, Milk, and BBQ Sauce may arise from inaccurate target effects, an implicit target representation that entangles object and task context, or competition between effect and visual-language tokens inside the action expert. Separating these explanations requires comparing effect error and action sensitivity on matched rescue and regression episodes.

TODO: Add the matched rescue/regression diagnostic proposed above, including effect-error correlations and representative qualitative rollouts; otherwise keep these failure explanations explicitly speculative.

V. CONCLUSION

We presented ArTE-VLA, a VLA architecture that predicts a compact physical future for the language-referred target and uses that prediction to condition action generation. The method requires privileged object states only for training-time supervision and retains the original RGB-language interface at inference. On LIBERO-Object, ArTE-VLA improves aggregate success from 70.9% to 84.3% over three evaluation seeds, while fixed-noise interventions show that the action expert responds consistently to changes in the predicted effects.

These results support a narrow conclusion: an autoregressive target-effect pathway can improve aggregate control performance and measurably influence the action computation. They do not yet show that future physical values are uniformly beneficial across tasks or solely responsible for the improvement. The regressions on several target objects and the learned horizon context carried by each effect token require matched auxiliary, capacity, and value-free controls before the contribution of explicit effect values can be isolated.

The method also relies on privileged target-effect labels during training. This supervision is compatible with simulation and instrumented robot datasets, but application to unconstrained real-world data would require state estimation or pseudo-labeling. Moreover, the implicit target representation does not expose a calibrated present-time object pose. An explicit target estimator could improve interpretability, but would introduce association and geometry errors that the current architecture avoids.

Finally, Table I uses one training seed, three stochastic evaluation seeds, and an expert-demonstration reset protocol. The repeated evaluations show that the aggregate gain is not specific to one rollout seed, but they do not establish robustness to training initialization or standard LIBERO initial states. Independent training-seed replication, completed

matched controls, standard-protocol evaluation, and tests beyond LIBERO-Object are therefore necessary to establish robustness and generalization. Within these limits, the results motivate structured future effects as an inspectable interface between visual-language understanding and continuous robot control.

TODO: Add standard LIBERO initial-state results, held-out task suites, and cross-simulator evaluation, or narrow the final generalization claims to the current LIBERO-Object expert-reset setting.

TODO: Revise the conclusion after the outstanding replications, ablations, and interventions; ensure that no final claim exceeds the completed evidence.

TODO: Complete the bibliography audit: expand coverage, replace arXiv entries with archival publication data where available, and verify author lists, titles, years, and identifiers.

TODO: Before submission, set `\draftnotesfalse`, confirm the anonymous author block for review, remove identifying metadata and comments, and rerun the PaperPlaza PDF compliance check.

REFERENCES

- [1] K. Kawaharazuka, J. Oh, J. Yamada, I. Posner, and Y. Zhu, "Vision-language-action models for robotics: A review towards real-world applications," *IEEE Access*, 2025.
- [2] X. Li, P. Li, M. Liu, D. Wang, J. Liu, B. Kang, X. Ma, T. Kong, H. Zhang, and H. Liu, "Towards generalist robot policies: What matters in building vision-language-action models," *arXiv preprint arXiv:2412.14058*, vol. 3, no. 7, p. 8, 2024.
- [3] Y. Ma, Z. Song, Y. Zhuang, J. Hao, and I. King, "A survey on vision-language-action models for embodied ai," *IEEE Transactions on Neural Networks and Learning Systems*, 2026.
- [4] R. Shao, W. Li, L. Zhang, R. Zhang, Z. Liu, R. Chen, and L. Nie, "Large vlm-based vision-language-action models for robotic manipulation: A survey," *arXiv preprint arXiv:2508.13073*, 2025.
- [5] Physical Intelligence *et al.*, " $\pi_{0.5}$: A vision-language-action model with open-world generalization," *arXiv preprint arXiv:2504.16054*, 2025.
- [6] M. U. Din, W. Akram, L. S. Saoud, J. Rosell, and I. Hussain, "Vision language action models in robotic manipulation: A systematic review," *arXiv preprint arXiv:2507.10672*, 2025.
- [7] R. Sapkota, Y. Cao, K. I. Roumeliotis, and M. Karkee, "Vision-language-action (vla) models: Concepts, progress, applications and challenges," *arXiv preprint arXiv:2505.04769*, 2025.
- [8] Y. Zhong, F. Bai, S. Cai, X. Huang, Z. Chen, X. Zhang, Y. Wang, S. Guo, T. Guan, K. N. Lui *et al.*, "A survey on vision-language-action models: An action tokenization perspective," *arXiv preprint arXiv:2507.01925*, 2025.
- [9] S. Wang, J. Shi, Z. Fu, X. He, F. Liu, C. Yang, Y. Zhou, Z. Fei, J. Gong, J. Fu *et al.*, "World action models: The next frontier in embodied ai," *arXiv preprint arXiv:2605.12090*, 2026.
- [10] W. Guo, Y. Wang, R. Zhou, Y. Zhang, R. Fang, Y. Pang, W. Xu, and G. Huang, "Action with visual primitives," *arXiv preprint arXiv:2605.22183*, 2026.
- [11] H. Xu, S. Zheng, H. Luo, W. Zhang, Z. Xi, and Z. Lu, "Unmasking the illusion of embodied reasoning in vision-language-action models," *arXiv preprint arXiv:2604.18000*, 2026.
- [12] Q. Zhao *et al.*, "CoT-VLA: Visual chain-of-thought reasoning for vision-language-action models," *arXiv preprint arXiv:2503.22020*, 2025.
- [13] J. Cen *et al.*, "WorldVLA: Towards autoregressive action world model," *arXiv preprint arXiv:2506.21539*, 2025.
- [14] C. Wen, X. Lin, J. So, K. Chen, Q. Dou, Y. Gao, and P. Abbeel, "Any-point trajectory modeling for policy learning," *arXiv preprint arXiv:2401.00025*, 2024.
- [15] H. Bharadhwaj, R. Mottaghi, A. Gupta, and S. Tulsiani, "Track2Act: Predicting point tracks from internet videos enables generalizable robot manipulation," *arXiv preprint arXiv:2405.01527*, 2024.
- [16] L. Zhong, Y. Liu, Y. Wei, Z. Xiong, S. Liu, and G. Ren, "Acot-vla: Action chain-of-thought for vision-language-action models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 8152–8162.
- [17] Octo Model Team *et al.*, "Octo: An open-source generalist robot policy," *arXiv preprint arXiv:2405.12213*, 2024.
- [18] M. J. Kim *et al.*, "OpenVLA: An open-source vision-language-action model," *arXiv preprint arXiv:2406.09246*, 2024.
- [19] K. Black *et al.*, " π_0 : A vision-language-action flow model for general robot control," *arXiv preprint arXiv:2410.24164*, 2024.
- [20] C.-L. Cheang *et al.*, "GR-2: A generative video-language-action model with web-scale knowledge for robot manipulation," *arXiv preprint arXiv:2410.06158*, 2024.
- [21] A. Stone *et al.*, "Open-world object manipulation using pre-trained vision-language models," *arXiv preprint arXiv:2303.00905*, 2023.
- [22] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, "LIBERO: Benchmarking knowledge transfer for lifelong robot learning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.